UNIVERSITY OF ILLINOIS URBANA-CHAMPAIGN

STAT 576: Empirical Process Theory and Weak Convergence Fall 2025
Lecturer: Eren C. Kızıldağ 10/09/2025
Scriber(s): Yangchang Liang

Lecture 14

Content.

1. Azuma's Inequality
2. Concentration for Discrete Lipschitz Functions (McDiarmid's Inequality)

1 Azuma's Inequality

Theorem 1 (Azuma's Inequality)**.** Let $(X_n)_{n\geq 0}$ be a martingale with $X_0 = 0$. Suppose $d_1, d_2, \ldots, d_n > 0$ such that

$$|X_i - X_{i-1}| \leq d_i, \quad i = 1, \ldots, n$$

almost surely. Then,

$$\mathbb{P}\left(|X_n| \geq t\right) \leq 2\exp\left(-\frac{t^2}{2\sum_{i=1}^n d_i^2}\right).$$

Remark: If $d_i = O(1)$, $\forall i$, then $\sum_i d_i^2 = O(n)$, so

$$\mathbb{P}\left(|X_n| \geq t\right) \lesssim \exp\left(-O\left(\frac{t^2}{n}\right)\right).$$

$$|X_n| \sim \Theta(\sqrt{n}).$$

In general, if X_n has a non-zero mean, this would imply:

$$\therefore \quad |X_n - \mathbb{E}X_n| = O(\sqrt{n}).$$

Proof. Recall that we proved last lecture:

$$f(x) \leq \frac{f(d_i) + f(-d_i)}{2} + \frac{f(d_i) - f(-d_i)}{2d_i}\, x,$$

for all $x \in [-d_i, d_i]$, where $f(x) = e^{\lambda x}$.

Next, note that:

$$\frac{f(d_i) + f(-d_i)}{2} = \frac{e^{\lambda d_i} + e^{-\lambda d_i}}{2} = \sum_{k=0}^{\infty} \frac{(\lambda d_i)^{2k}}{(2k)!} \leq \sum_{k=0}^{\infty} \frac{(\lambda^2 d_i^2/2)^k}{k!} = \exp\left(\frac{\lambda^2 d_i^2}{2}\right).$$

where we used the fact that

$$(2k)! \geq 2^k k!$$

So,

$$f(x) \leq \exp\left(\frac{\lambda^2 d_i^2}{2}\right) + \frac{f(d_i) - f(-d_i)}{2d_i}\, x.$$

We now proceed analogously to the proof of Hoeffding's inequality.

$$\begin{aligned}\mathbb{P}[X_n \geq t] &= \mathbb{P}[e^{\lambda X_n} \geq e^{\lambda t}], \quad \lambda \geq 0, \\ &\leq e^{-\lambda t}\, \mathbb{E}[e^{\lambda X_n}].\end{aligned}$$

We now control the expectation above.

$$X_n = \sum_{i=1}^{n} (X_i - X_{i-1}).$$

Claim: $\mathbb{E}[X_i - X_{i-1}] = 0.$

Proof.

$$\mathbb{E}[X_i \mid \mathcal{F}_{i-1}] = X_{i-1} \quad \text{(Martingale)}.$$

$$\mathbb{E}[X_i - X_{i-1}] = \mathbb{E}\big[\mathbb{E}[(X_i - X_{i-1}) \mid \mathcal{F}_{i-1}]\big] = \mathbb{E}[X_{i-1} - X_{i-1}] = 0.$$

□

Now,

$$\begin{aligned}\mathbb{E}\left[\exp\left(\lambda \sum_{i=1}^{n} (X_i - X_{i-1})\right)\right] &= \mathbb{E}\left[\mathbb{E}\left[\exp\left(\lambda \sum_{i=1}^{n} (X_i - X_{i-1})\right) \Big| \mathcal{F}_{n-1}\right]\right] \quad \text{(tower property)} \\ &= \mathbb{E}\left[\exp\left(\sum_{i=1}^{n-1} \lambda (X_i - X_{i-1})\right) \mathbb{E}\left[e^{\lambda (X_n - X_{n-1})} \Big| \mathcal{F}_{n-1}\right]\right],\end{aligned}$$

$$(\text{since } \mathbb{E}\big[\mathbb{E}[f(X)g(Y) \mid Y]\big] = \mathbb{E}\big[g(Y)\mathbb{E}[f(X) \mid Y]\big]).$$

Since $|X_n - X_{n-1}| \le d_n$ a.s., we have

$$e^{\lambda(X_n - X_{n-1})} \le \exp\left(\frac{\lambda^2 d_n^2}{2}\right) + [\text{const}](X_n - X_{n-1}),$$

and hence

$$\mathbb{E}\Big[e^{\lambda(X_n - X_{n-1})}\Big|\mathcal{F}_{n-1}\Big] \le \exp\left(\frac{\lambda^2 d_n^2}{2}\right) + [\text{const}]\,\mathbb{E}[X_n - X_{n-1} \mid \mathcal{F}_{n-1}]\,.$$

Because $\mathbb{E}[X_n - X_{n-1} \mid \mathcal{F}_{n-1}] = 0$ (martingale property),

$$\mathbb{E}\Big[e^{\lambda(X_n - X_{n-1})}\Big|\mathcal{F}_{n-1}\Big] = \exp\left(\frac{\lambda^2 d_n^2}{2}\right).$$

Thus,

$$\mathbb{E}\left[\exp\left(\lambda\sum_{i=1}^{n}(X_i - X_{i-1})\right)\right] \le \exp\left(\frac{\lambda^2 d_n^2}{2}\right)\mathbb{E}\left[\exp\left(\lambda\sum_{i=1}^{n-1}(X_i - X_{i-1})\right)\right].$$

$$\vdots$$

$$\le \exp\left(\frac{\lambda^2}{2}\sum_{i=1}^{n} d_i^2\right), \quad \text{by induction.}$$

Therefore,

$$\mathbb{P}[X_n \ge t] \le \exp\left(-\lambda t + \frac{\lambda^2}{2}\sum_{i=1}^{n} d_i^2\right) =: \exp\big(\phi(\lambda)\big).$$

$$\phi'(\lambda) = -t + \lambda\sum_{i=1}^{n} d_i^2 \quad \Rightarrow \quad \phi(\lambda) \ge \phi(\lambda^*), \quad \lambda^* = \frac{t}{\sum_{i=1}^{n} d_i^2}.$$

Plugging this value establishes Azuma's inequality. □

2 Concentration for Discrete Lipschitz Functions

Setup: Let $f : \mathbb{R}^n \to \mathbb{R}$ be discrete Lipschitz. That is, there exist constants $d_1, \ldots, d_n$ such that

$$\left|f(x_1, \ldots, x_i, \ldots, x_n) - f(x_1, \ldots, x_i', \ldots, x_n)\right| \leq d_i,$$

for all i, and for all $x_1, \ldots, x_{i-1}, x_i, x_i', x_{i+1} \ldots, x_n$.

Equivalently,

$$|f(x) - f(y)| \leq \sum_{i=1}^{n} d_i \, \mathbf{1}\{x_i \neq y_i\}, \quad x, y \in \mathbb{R}^n.$$

Exercise: Show by triangle inequality that both versions of the discrete Lipschitz condition are equivalent.

Theorem 2 (McDiarmid's inequality)**.** Suppose that $X_1, \ldots, X_n$ are independent random variables and f be discrete Lipschitz with constants $d_1, \ldots, d_n$. Then, assuming $\mathbb{E}|f| < \infty$,

$$\mathbb{P}[\,|f(X_1, \ldots, X_n) - \mathbb{E}f(X_1, \ldots, X_n)| \geq t\,] \leq 2\exp\left(-\frac{t^2}{2\sum_{i=1}^n d_i^2}\right).$$

Proof. Let's define filtrations:

$$\mathcal{F}_0 = \{\emptyset, \Omega\}, \quad \mathcal{F}_1 = \sigma(X_1), \quad \mathcal{F}_2 = \sigma(X_1, X_2), \ \ldots$$

Clearly, $\mathcal{F}_0 \subset \mathcal{F}_1 \subset \mathcal{F}_2 \subset \cdots \subset \mathcal{F}_n$.

Define the Doob martingale:

$$M_i := \mathbb{E}[f(X_1, \ldots, X_n) \mid X_1, \ldots, X_i] = \mathbb{E}[f(X_1, \ldots, X_n) \mid \mathcal{F}_i]\,.$$

$$M_n = f(X_1, \ldots, X_n), \quad M_0 = \mathbb{E}f(X_1, \ldots, X_n).$$

Note that $(M_i - M_0)_{i \geq 0}$ is a martingale.

$$M_n - M_0 = f(X_1, \ldots, X_n) - \mathbb{E}f(X_1, \ldots, X_n) = \sum_{i=1}^{n} (M_i - M_{i-1}).$$

Note that if $|M_i - M_{i-1}| \leq d_i$, we are done by Azuma.

Lemma 1. For any i, we have

$$|M_i - M_{i-1}| \leq d_i \quad \text{almost surely.}$$

Proof of Lemma.

$$M_i = \mathbb{E}[f(X_1, \ldots, X_n) \mid X_1, \ldots, X_i] = \int f(x_1, \ldots, x_i, x_{i+1}, \ldots, x_n)\, d\mathbb{P}(x_{i+1}) \cdots d\mathbb{P}(x_n),$$

(this step uses independence).

$$M_{i-1} = \int f(X_1, \ldots, X_{i-1}, x_i, x_{i+1}, \ldots, x_n)\, d\mathbb{P}(x_i)\, d\mathbb{P}(x_{i+1}) \cdots d\mathbb{P}(x_n).$$

By Fubini–Tonelli,

$$\int \left(\int f(X_1, \ldots, X_{i-1}, x_i, x_{i+1}, \ldots, x_n)\, d\mathbb{P}(x_i) \right) d\mathbb{P}(x_{i+1}) \cdots d\mathbb{P}(x_n).$$

Thus,

$$|M_i - M_{i-1}| \le \int \left| \int \Big(f(X_1, \ldots, X_i, x_{i+1}, \ldots, x_n) - f(X_1, \ldots, X_{i-1}, x_i, x_{i+1}, \ldots, x_n) \Big) d\mathbb{P}(x_i) \right| d\mathbb{P}(x_{i+1}) \cdots d\mathbb{P}(x_n)$$

$$\Big| f(x_1, \ldots, x_i, x_{i+1}, \ldots, x_n) - \int f(x_1, \ldots, x_{i-1}, x_i', x_{i+1}, \ldots, x_n)\, d\mathbb{P}(x_i') \Big| \le d_i.$$

Hence,

$$|M_i - M_{i-1}| \le d_i, \quad \text{a.s.}$$

□

Thus, Azuma's inequality yields the conclusion.

□

Applications

(1) Concentration of Empirical Means

Let $X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} F$. Define the empirical CDF:

$$F_n(x) = \frac{1}{n} \sum_{i=1}^{n} \mathbf{1}\{X_i \le x\}.$$

Define the Kolmogorov–Smirnov statistic:

$$L_n := \sup_{x \in \mathbb{R}} \big| F_n(x) - F(x) \big|.$$

Theorem 3 (Dvoretzky–Kiefer–Wolfowitz)**.**

$$\mathbb{P}[\,|L_n - \mathbb{E}L_n| \geq t\,] \leq 2\exp\left(-\frac{t^2 n}{2}\right).$$

Proof. Recall that

$$L_n := L_n(X_1, \ldots, X_n),$$

and note that

$$|L_n(X_1, \ldots, X_i, \ldots, X_n) - L_n(X_1, \ldots, X_i', \ldots, X_n)| \leq \frac{1}{n}.$$

By McDiarmid's inequality,

$$\Pr[\,|L_n - \mathbb{E}L_n| \geq t\,] \leq 2\exp\left(-\frac{t^2}{2\sum_{i=1}^n (1/n)^2}\right) = 2\exp\left(-\frac{t^2 n}{2}\right).$$

□

(2) U-Statistics

Let $X_1, \ldots, X_n \sim F$.

Goal: Estimate $\mathbb{E}|Y - Z|$, where $Y, Z \overset{\text{i.i.d.}}{\sim} F$.

$$f(X_1, \ldots, X_n) := \frac{1}{\binom{n}{2}} \sum_{1 \leq i < j \leq n} |X_i - X_j|.$$

Check that

$$\mathbb{E}\, f(X_1, \ldots, X_n) = \mathbb{E}|Y - Z|.$$

Assume X_i's have bounded support $[-a, a]$.

$$\big|f(X_1, \ldots, X_i, \ldots, X_n) - f(X_1, \ldots, X_i', \ldots, X_n)\big| = \left|\frac{1}{\binom{n}{2}} \sum_{1 \leq i < j \leq n} |X_j - X_i| - \frac{1}{\binom{n}{2}} \sum_{1 \leq i < j \leq n} |X_j - X_i'|\right|$$

$$\leq \frac{1}{\binom{n}{2}} \sum_{1 \leq i < j \leq n} \big|\,|X_j - X_i| - |X_j - X_i'|\,\big|.$$

Reverse Δ-Inequality

$$\big|\,|x| - |y|\,\big| \leq |x - y|.$$

Therefore,

$$\big|f(X_1,\ldots,X_i,\ldots,X_n) - f(X_1,\ldots,X_i',\ldots,X_n)\big| \le \frac{1}{\binom{n}{2}} \sum_{1\le i<j\le n} |X_i - X_i'| \le \frac{1}{\binom{n}{2}}(n-1)2a = \frac{4a}{n}.$$

Thus,

$$\mathbb{P}[\,|f - \mathbb{E}|Y-Z|| \ge t\,] \le 2\exp\left(-\frac{t^2}{2\sum_i (4a/n)^2}\right) = 2\exp\left(-\frac{t^2 n}{32a^2}\right).$$

If X_i had unbounded support: Let

$$E_t := \{\,|X_i| \le t\,\}.$$

Then,

$$\begin{aligned}\mathbb{P}[\,|f - \mathbb{E}|Y-Z|\,| \ge t\,] \le & \mathbb{P}[\,|f - \mathbb{E}|Y-Z|| \ge t \,|\, E_{t_1},\ldots,E_{t_n}\,]\,\mathbb{P}[E_{t_1},\ldots,E_{t_n}] \\ & + \mathbb{P}[\,|f - \mathbb{E}|Y-Z|| \ge t \,|(E_{t_1}\cap\cdots\cap E_{t_n})^c]\,\mathbb{P}[(E_{t_1}\cap\cdots\cap E_{t_n})^c].\end{aligned}$$